

Lecture 07: Transposition Cipher

Transposition Cipher

A Transposition Cipher, also known as a Permutation Cipher, is a classical encryption technique in which the positions of the letters are changed, but the letters themselves remain unchanged. Substitution ciphers, where one letter is replaced by another, transposition ciphers simply rearrange the order of characters to produce the ciphertext. Because the original letters remain the same, the frequency distribution of letters does not change, making transposition ciphers different from substitution ciphers.

Basic Idea

Suppose the plaintext is

HELLO

A substitution cipher might change it to

KHOOR

where letters themselves change.

A transposition cipher might produce

LEHOL

Here,

- Letters remain exactly the same.
- Only their positions change.

Characteristics of Transposition Cipher

- Rearranges letter positions.
- Does not replace characters.
- Plaintext and ciphertext contain exactly the same letters.
- Letter frequency remains unchanged.
- Also called a **Permutation Cipher**.

Working Principle

Encryption process:

Plaintext

|



Arrange letters according to a rule

|



Rearrange positions

|



Ciphertext

Decryption simply performs the reverse arrangement.

Types of Transposition Ciphers

There are two important transposition ciphers:

- Rail Fence Cipher
- Columnar Transposition Cipher

Rail Fence Cipher

The Rail Fence Cipher is one of the simplest transposition ciphers. In this method, the plaintext is written diagonally in a zig-zag pattern across multiple rows called rails. After writing all letters, the rows are read one by one to form the ciphertext.

Basic Idea

Imagine several horizontal railway tracks.

The message is written by moving:

Down

↓

Then

Up

↑

Then

Down again

forming a zig-zag.

Finally, Read row by row.

Rail Fence Encryption Steps

Step 1

Choose the number of rails (key).

Example

Key = 3

Step 2

Write the plaintext diagonally.

Move

Top → Bottom

then

Bottom → Top

continuously.

Step 3

After all letters are written,

Read every row from left to right.

The combined rows form the ciphertext.

Rail Fence Example

D				N				Y		
	E		O		S		A		E	
		M				L				R

Plaintext
DEMONSLAYER

Key
3 Rails
Arrange in zig-zag
Rail 1: D N Y

Rail 2: E O S A E

Rail 3: M L R
Now read rows
DNY

EOSAE

MLR
Ciphertext
DNYEOSAEMLR

Rail Fence Decryption

Decryption is the reverse of encryption.

Steps

Step 1

Find

- Length of ciphertext
- Number of rails (key)

Step 2

Create an empty zig-zag matrix.

Step 3

Mark the zig-zag positions.

Step 4

Fill the ciphertext row by row into the marked positions.

Step 5

Read the matrix diagonally in zig-zag order.
The original plaintext is obtained.

Example

Ciphertext

DNYEOSAEMLR

Key

3 Rails

Step 1: Mark the zig-zag positions

Rail 1: * * *

Rail 2: * * * * *

Rail 3: * * *

Step 2: Fill the ciphertext row by row

Rail 1 gets the first 3 letters:

Rail 1: D N Y

Rail 2 gets the next 5 letters:

Rail 2: E O S A E

Rail 3 gets the last 3 letters:

Rail 3: M L R

So the matrix becomes:

Rail 1: D N Y

Rail 2: E O S A E

Rail 3: M L R

Step 3: Read in zig-zag order

D → E → M → O → N → S → L → A → Y → E → R

Plaintext

DEMONSLAYER

This matches the format of your encryption example and is easy to follow for exams.

Advantages of Rail Fence Cipher

- Very easy to understand
- Simple implementation
- Fast encryption
- No letter substitution
- Useful for learning classical cryptography

Disadvantages

- Very weak security
- Easily broken using brute force
- Letter frequency remains unchanged
- Not suitable for modern communication

Columnar Transposition Cipher

Columnar Transposition Cipher is a classical encryption technique in which the plaintext is written into a table row by row under a keyword, and the columns are rearranged and read in a specific order determined by the alphabetical order of the keyword to produce the ciphertext.

Basic Idea

Write message

↓

Inside rows

↓

Assign numbers using keyword

↓

Read columns

↓

Ciphertext

Keyword

The keyword determines

- Number of columns
- Reading order

Example

G	U	I	T	A	R
2	6	3	5	1	4
B	o	h	e	m	i
a	n		R	h	a
p	s	o	d	y	-

Plaintext: Bohemian Rhapsody

Keyword: GUITAR (length=6)

Arrange alphabetically

A G I R T U

Now assign numbers

G U I T A R

2 6 3 5 1 4

Columnar Encryption Steps

Step 1

Choose a keyword.

Example

GUITAR

Step 2

Number the keyword alphabetically.

G U I T A R

2 6 3 5 1 4

Step 3

Write the plaintext row by row.

Example

Bohemian Rhapsody

G U I T A R

B o h e m i

a n _ R h a

p s o d y -

The "-" (dash) is used in the last box because the plaintext length is not enough to completely fill the table in the Columnar Transposition Cipher.

Step 4

Read columns according to the assigned numbers.

Result: mhyBaphoia-eRdons

This becomes the ciphertext.

Columnar Decryption

Steps

Step 1

Determine

- Key length
- Number of rows

Step 2

Write ciphertext column by column.

Step 3

Restore columns according to the original keyword order.

Step 4

Read row by row.

Original plaintext is recovered.

Example

Ciphertext

mhyBaphoia-eRdons

Keyword

G U I T A R

2 6 3 5 1 4

Step 1

Key length = 6

Ciphertext length = 17

Number of rows = 3

Step 2

Fill the ciphertext column by column according to the column numbers.

Column 1 (A): m h y

Column 2 (G): B a p

Column 3 (I): h o i

Column 4 (R): a - e

Column 5 (T): R d o

Column 6 (U): n s

Step 3

Restore the columns to the original keyword order.

G U I T A R

B o h e m i

a n - R h a

p s o d y -

Step 4

Read row by row.

Bohemian Rhapsody

Plaintext

Bohemian Rhapsody

Note: If the last row is incomplete, empty cells are left blank or filled with a padding character (such as -) during encryption and ignored during decryption.

Advantages of Columnar Transposition

- More secure than Rail Fence Cipher
- Simple algorithm
- Easy implementation
- Maintains original letters

Disadvantages

- Frequency analysis still possible
- Weak against modern attacks
- Depends on keeping the keyword secret

Rail Fence vs Columnar Transposition

Feature	Rail Fence	Columnar
Arrangement	Zig-zag	Table
Key	Number of rails	Keyword
Complexity	Simple	Moderate
Security	Low	Better
Matrix	Zig-zag matrix	Row-column matrix

Product Cipher

A Product Cipher is a cryptographic technique that combines substitution and transposition ciphers to produce stronger encryption. Using only substitution or only transposition is not secure enough because language characteristics can still reveal patterns. By combining both methods, the resulting cipher becomes much harder to break. This concept forms the bridge between classical and modern cryptography.

Why Product Cipher?

Suppose

Only Substitution

↓

Still vulnerable.

Only Transposition

↓

Still vulnerable.

But

Substitution

↓

Transposition

↓

Produces a much stronger cipher.

Important Facts

- Two substitutions create a more complex substitution.
- Two transpositions create a more complex transposition.
- A substitution followed by a transposition creates a significantly stronger cipher.

Advantages

- Higher security
- More resistant to attacks
- Foundation of modern encryption algorithms

Rotor Machines

Before modern computer-based encryption, Rotor Machines were electromechanical devices used to perform complex substitution ciphers. They became famous during World War II, especially through the German Enigma Machine. Other examples include the Hagelin and Japanese Purple machines.

Working Principle

Rotor machines contain several rotating cylinders (rotors). Each rotor performs a substitution.

After encrypting one letter,

- Rotor rotates.
- Substitution changes.
- Next letter is encrypted differently.

Therefore, the same plaintext letter may encrypt into different ciphertext letters depending on its position.

Encryption Process

Letter

↓

Rotor 1

↓

Rotor 2

↓
Rotor 3
↓
Cipher Letter
↓
Rotors Rotate
↓
Next Letter Uses Different Mapping

Advantages

- Very strong for its time
- Dynamic substitution
- Huge number of possible keys

Disadvantages

- Large mechanical device
- Slow compared to computers
- Eventually broken through cryptanalysis

Examples

- German Enigma
- Hagelin Machine
- Japanese Purple

Steganography

Steganography is the art of hiding the existence of a secret message inside another ordinary object, such as an image, audio file, video, or text. Unlike encryption, which hides the contents of a message, steganography hides the fact that a secret message even exists.

Encryption vs Steganography

Encryption
Visible message
↓
Unreadable

Steganography
Message
↓
Hidden
↓
Nobody knows it exists

Common Methods

1. Invisible Ink

Hidden writing that becomes visible only under special conditions.

2. Hidden Text

Secret letters are embedded inside a larger message.

3. Image Steganography

Secret bits are hidden inside image pixels.

4. Audio Steganography

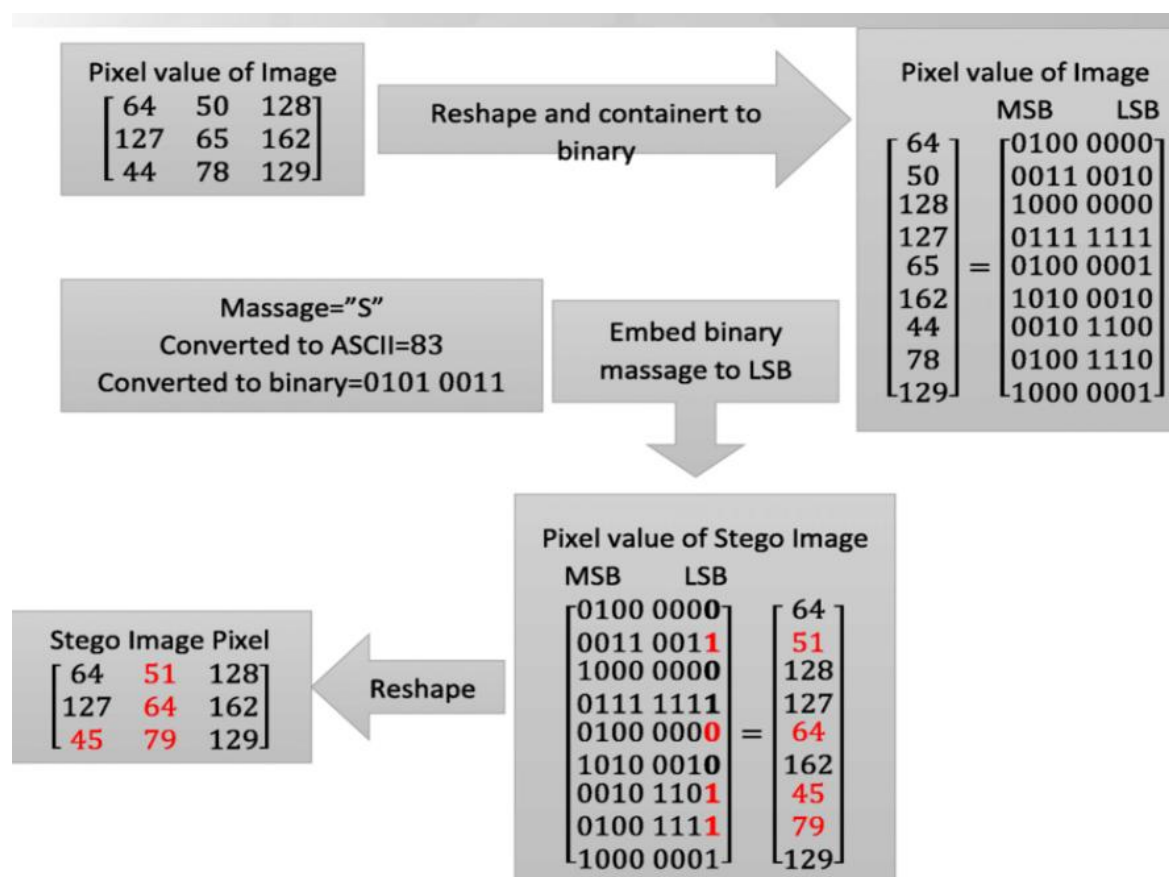
Hidden inside audio samples.

5. Video Steganography

Hidden inside video frames.

LSB (Least Significant Bit) Steganography

The most common steganography technique is Least Significant Bit (LSB) embedding. Each image pixel is represented by binary bits. The least significant bit of each pixel is replaced with one bit of the secret message. Since only the last bit changes, the visual appearance of the image remains almost identical to the original. The lecture demonstrates embedding the binary representation of the character "S" (ASCII 83) into the LSBs of image pixels.



LSB Process

Original Image

↓

Convert pixels to binary

↓

Convert secret message to binary

↓

Replace each pixel's last bit

↓

Stego Image

Advantages of Steganography

- Secret communication
- Message existence remains hidden
- Can be combined with encryption
- Difficult to detect by ordinary users

Disadvantages

- Low data capacity
- Large files needed
- Can be destroyed by compression or editing
- Specialized detection tools may reveal hidden data

Comparison: Encryption vs Steganography

Feature	Encryption	Steganography
Purpose	Hide message content	Hide message existence
Visibility	Message is visible but unreadable	Message is invisible
Security	Mathematical algorithms	Hidden storage
Example	AES, RSA	LSB Image Hiding